

Operational Diagnostics Report

Nairobi Throughput Shortfall — Root Cause & Recommendations | Prepared by Stanley Metone | Week 5

Context

Across our four depots, daily throughput should sit consistently in the 950–1,050 barrel range. A review of 2026 production records shows that is **not** happening everywhere. Nairobi's output is erratic — some days near target, others far below it — and this pattern has been present for nearly the entire year. We investigated the full dataset (2,920 daily records, 13 machines, 4 depots) to find out why.

Insight: What We Found

The problem is one machine: NBI-P03 in Nairobi. Since **February 2026**, this pump has skipped its scheduled maintenance on over 90% of the days it operated — every month since, without recovery. On those days, its output falls to roughly **625 barrels**, down from a healthy ~990–1,000 barrels — a 37% shortfall. No other machine in the company shows this pattern at all.

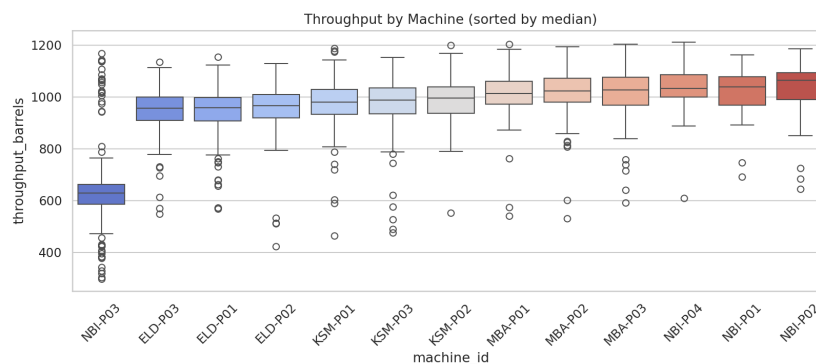


Figure 1: Daily throughput by machine. NBI-P03 sits well below every other unit fleet-wide, and its wide spread shows it swinging between normal and depressed days.

We ruled out the obvious alternative explanations before settling on this conclusion:

- **Not the weather.** Ambient temperature has almost no relationship with throughput (correlation of 0.25 out of a possible 1.0).
- **Not the power supply.** Voltage readings show essentially zero relationship with throughput (correlation of 0.04).
- **Not one careless operator.** All five operators rostered on NBI-P03 show nearly identical missed-maintenance rates (92–95%) — this is a machine/process issue, not a person issue.
- **It is maintenance.** Skipped maintenance has a strong negative correlation with throughput (–0.85) — by far the strongest relationship in the data.

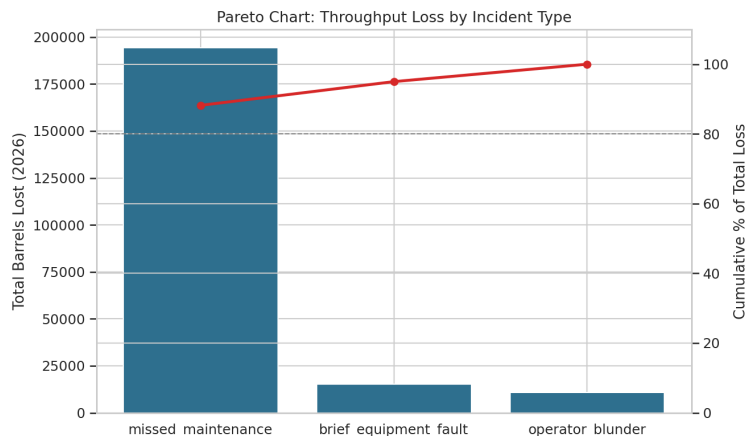


Figure 2: Pareto chart of throughput lost to incidents in 2026. Missed maintenance on NBI-P03 accounts for ~88% of all incident-driven barrel losses company-wide — equipment faults and operator errors are minor by comparison.

In barrels, missed maintenance on this single machine cost roughly **194,000 barrels** of lost production in 2026 — nearly nine times more than equipment faults and operator errors combined.

Action: Recommendations

#	Recommendation	Why it matters
1	Audit and reset NBI-P03's maintenance schedule this month. Assign a named owner accountable for sign-off after every service window, not just the technician on shift.	The failure is chronic and machine-specific, not a one-off skip — it needs a hard reset, not a reminder.
2	Add a simple weekly maintenance-compliance report per machine (% of scheduled services completed on time), reviewed at the depot level.	This failure ran undetected for 10+ months. A lightweight recurring check would have caught it in weeks, not a year-end review.
3	Once NBI-P03 is back on schedule, re-baseline its expected throughput and monitor for 60 days to confirm output returns to the ~990–1,000 barrel/day range seen at other depots.	Confirms the fix worked and gives an early warning if the issue recurs.

Supporting Evidence

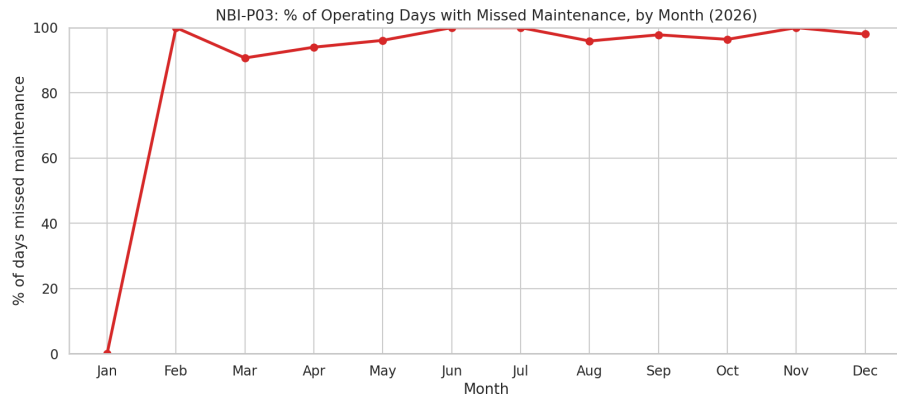


Figure 3: Share of NBI-P03's operating days with a missed-maintenance flag, by month. January was clean; the issue appears in February and never recovers through December — confirming this is an ongoing process failure, not a single incident.

Bottom line: a fixable, process-level issue on one asset — not weather, power, or people. Restoring NBI-P03's maintenance discipline is the single highest-leverage action available this quarter.